

Windows Server and Network Infrastructure Home Lab

Project Documentation | Active Directory, DNS, Group Policy, File Services

Item	Details
Author	KYLA TRISHA B. TOLIAO
Date completed	OCTOBER 4, 2026
Repository	https://github.com/ktbtoliao-gif/active-directory-home-lab
Environment	VMware Workstation, Windows Server 2022, Windows 10
Domain	lab.local (NetBIOS name: LAB)

1. Project overview

This lab simulates a small-business Windows environment: a domain controller that provides identity, DNS, Group Policy, and file services, and a domain-joined workstation. It was built to practice the daily work of a system administrator and network support technician, and every configuration below was verified with a test.

Objectives

- Design an isolated IPv4 network with static addressing and DNS that supports Active Directory
- Deploy and verify a Windows Server 2022 domain controller and join a Windows 10 client
- Organize users and computers with OUs and security groups
- Enforce policy through Group Policy (password and lockout rules, restrictions, mapped drives, banner, auditing)
- Secure department file shares with group-based NTFS permissions (least privilege)
- Practice troubleshooting with command-line tools and Windows event logs

Tools and versions

Component	Details
Hypervisor	VMware Workstation Pro 25H2u1
Host machine	Windows 11 Home, 16 GB RAM, 12 th Gen Intel Core i5-12500H
Server VM	Windows Server 2022 Datacenter Evaluation, 4 GB RAM, about 60 GB disk, hostname WIN2022DC
Client VM	Windows 10 version 22H2 (build 19045), 4 GB RAM, 60 GB disk
Network	VMnet1 (host-only), 192.168.10.0/24

2. Lab architecture

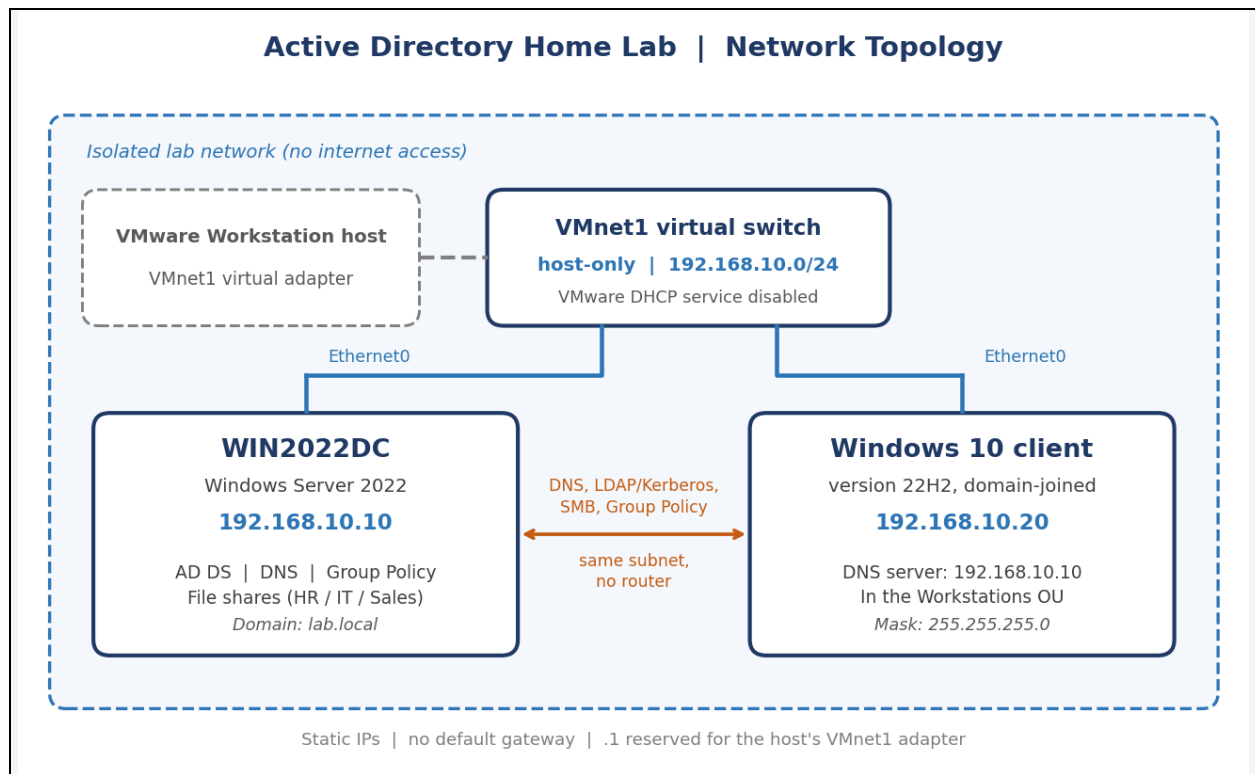


Figure 1. Network diagram of the lab

Machine	Role	IP address	Mask	Gateway	DNS server
WIN2022DC	Domain controller, DNS, file server	192.168.10.10	255.255.255.0	None	127.0.0.1 (itself)
Windows 10 client	Domain-joined workstation	192.168.10.20	255.255.255.0	None	192.168.10.10

Design decisions

- Private range 192.168.10.0/24 (RFC 1918): 254 usable hosts, with .0 as the network address and .255 as the broadcast address.
- Same subnet and mask on both machines, so each treats the other as local and sends traffic directly without a router.
- No default gateway, because the host-only network is isolated and has nothing outside the subnet to reach.
- Static addresses with VMware's DHCP service disabled, because the server and the domain depend on a fixed address. The .1 address was avoided because the host's VMnet1 adapter uses it.
- The client's DNS points to the domain controller, because Active Directory locates domain controllers through DNS SRV records.

3. Build and verification

3.1 VMware virtual network

Objective: put both VMs on one isolated virtual network that I fully control.

1. Opened Edit > Virtual Network Editor and selected VMnet1 (host-only).
2. Unticked "Use local DHCP service" so VMware would not hand out addresses.
3. Confirmed the subnet is 192.168.10.0 with mask 255.255.255.0.
4. On each VM, set the network adapter to VMnet1 with Connected and Connect at power on ticked.

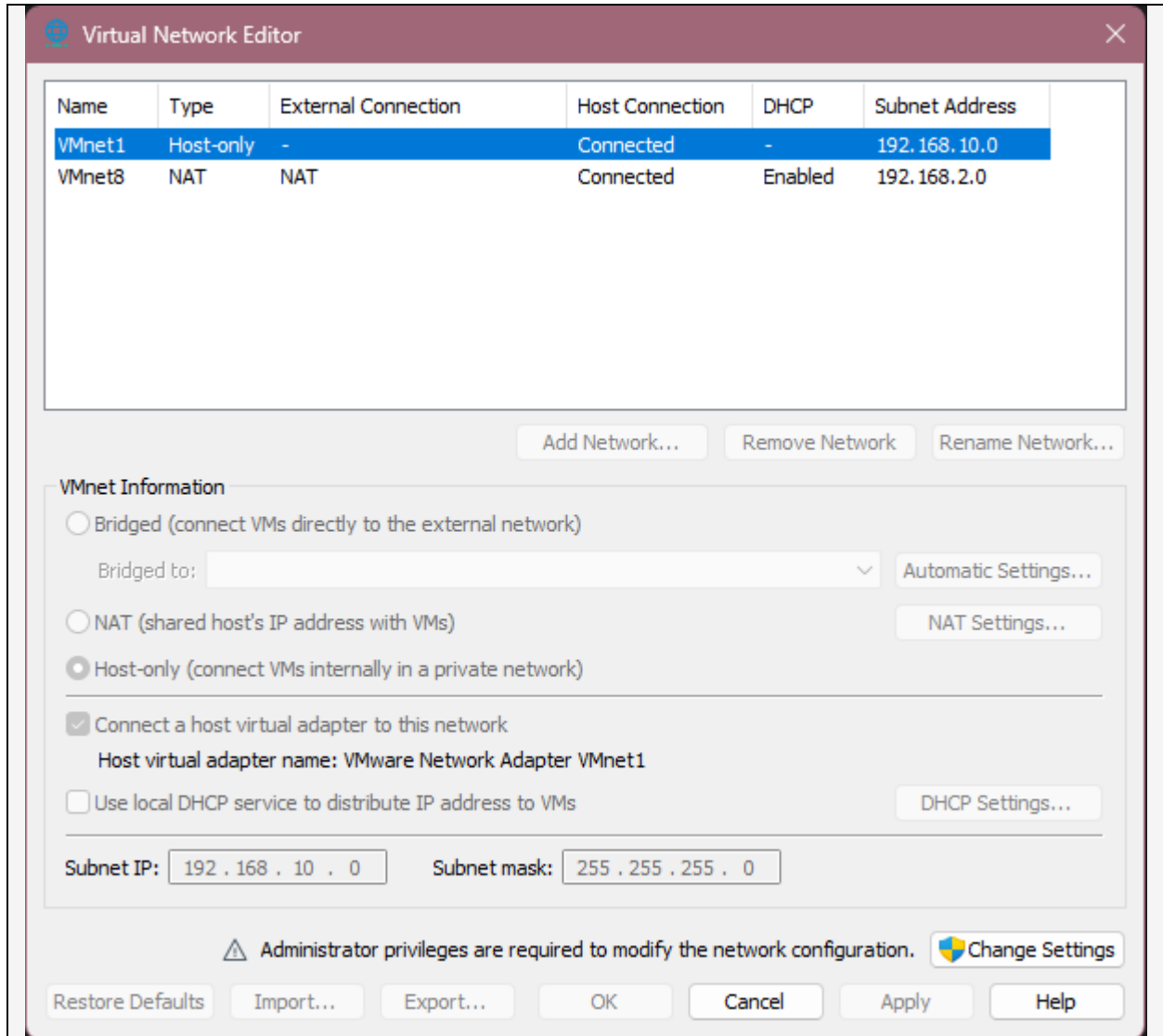


Figure 2. Virtual Network Editor showing VMnet1 settings

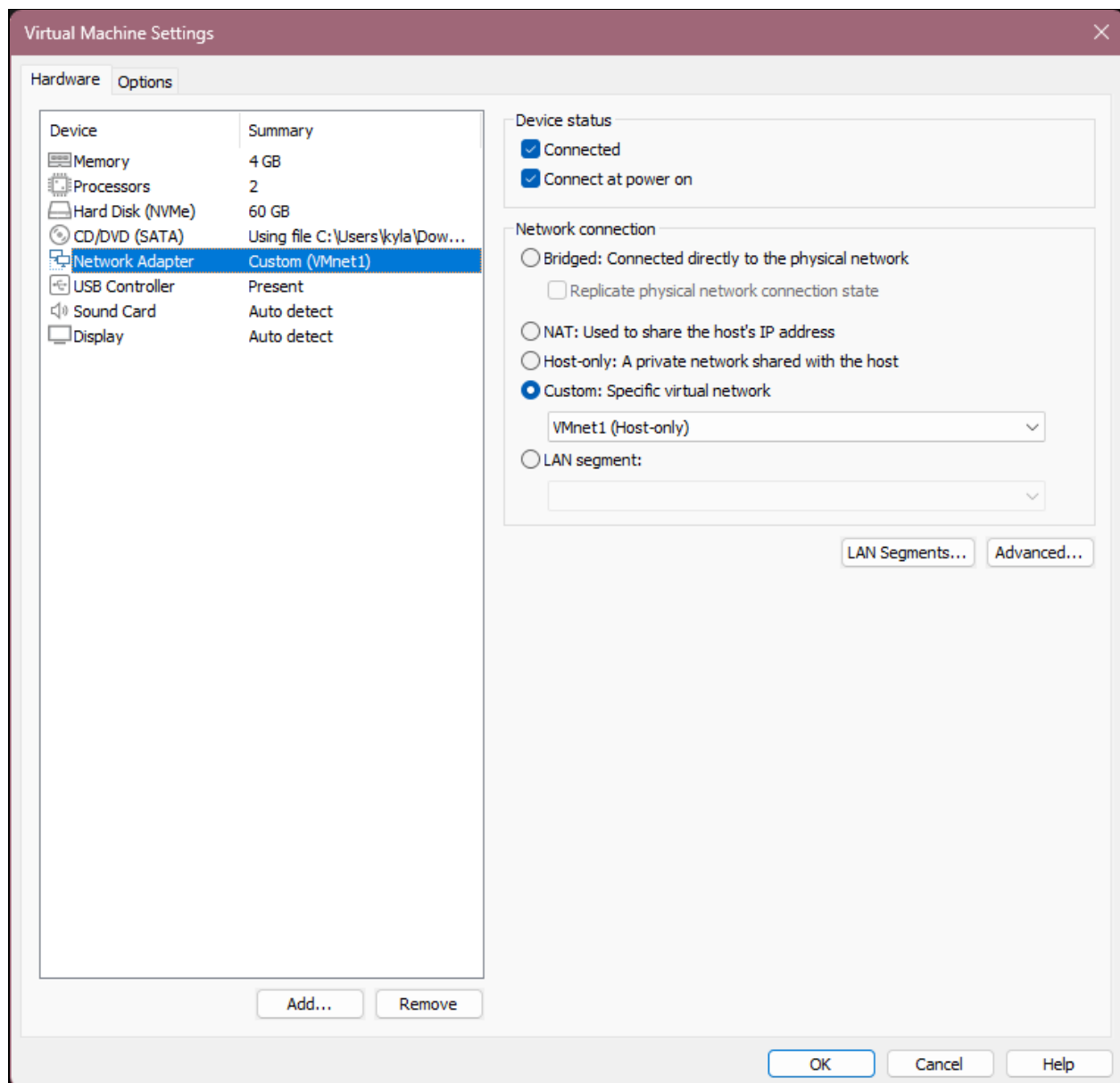


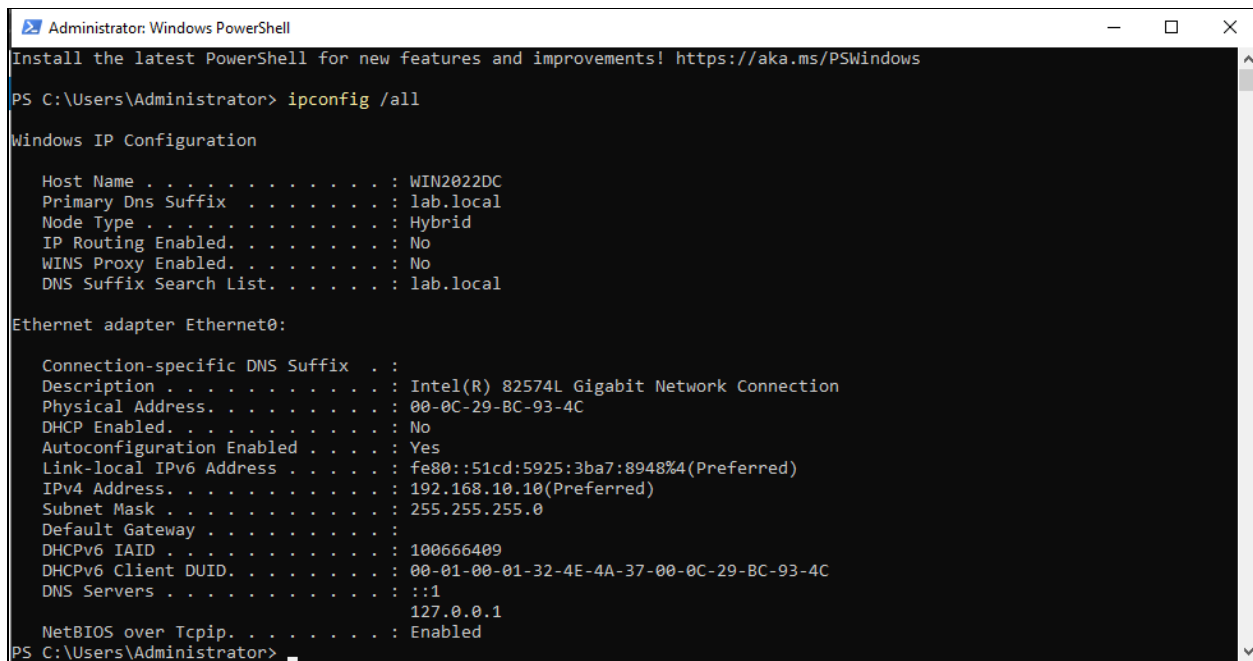
Figure 3. VM network adapter settings (one VM is enough)

3.2 Static IP addressing

Objective: give each machine a fixed address in the same subnet.

1. On each VM, opened `ncpa.cpl` > Ethernet0 > Properties > Internet Protocol Version 4.
2. Entered the address, mask, and DNS server from the addressing table in section 2, leaving the gateway blank.
3. Verified the configuration with `ipconfig /all`.

```
ipconfig /all
```



```

Administrator: Windows PowerShell
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : WIN2022DC
    Primary Dns Suffix . . . . . : lab.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : lab.local

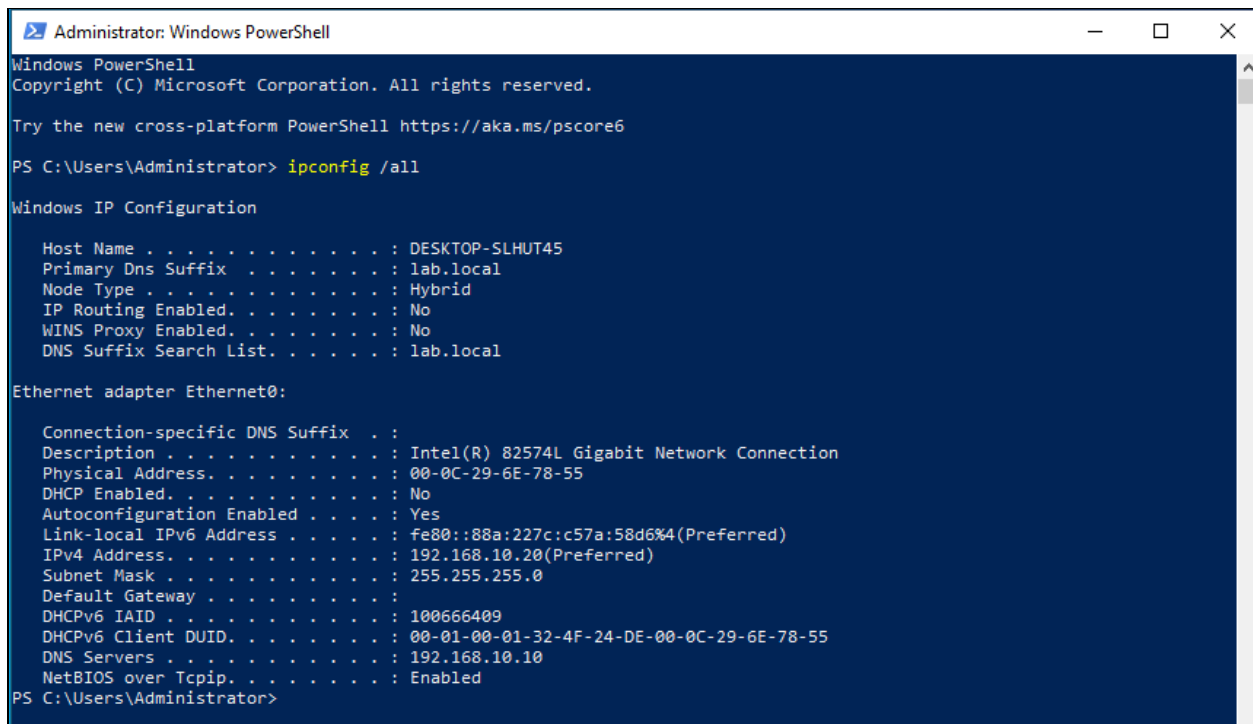
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-BC-93-4C
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::51cd:5925:3ba7:8948%4(Preferred)
    IPv4 Address. . . . . : 192.168.10.10(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-4E-4A-37-00-0C-29-BC-93-4C
    DNS Servers . . . . . : ::1
                           127.0.0.1
    NetBIOS over Tcpip. . . . . : Enabled

PS C:\Users\Administrator>

```

Figure 4. Server IP configuration



```

Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\Users\Administrator> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-SLHUT45
    Primary Dns Suffix . . . . . : lab.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : lab.local

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-6E-78-55
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::88a:227c:c57a:58d6%4(Preferred)
    IPv4 Address. . . . . : 192.168.10.20(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 100666409
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-4F-24-DE-00-0C-29-6E-78-55
    DNS Servers . . . . . : 192.168.10.10
    NetBIOS over Tcpip. . . . . : Enabled

PS C:\Users\Administrator>

```

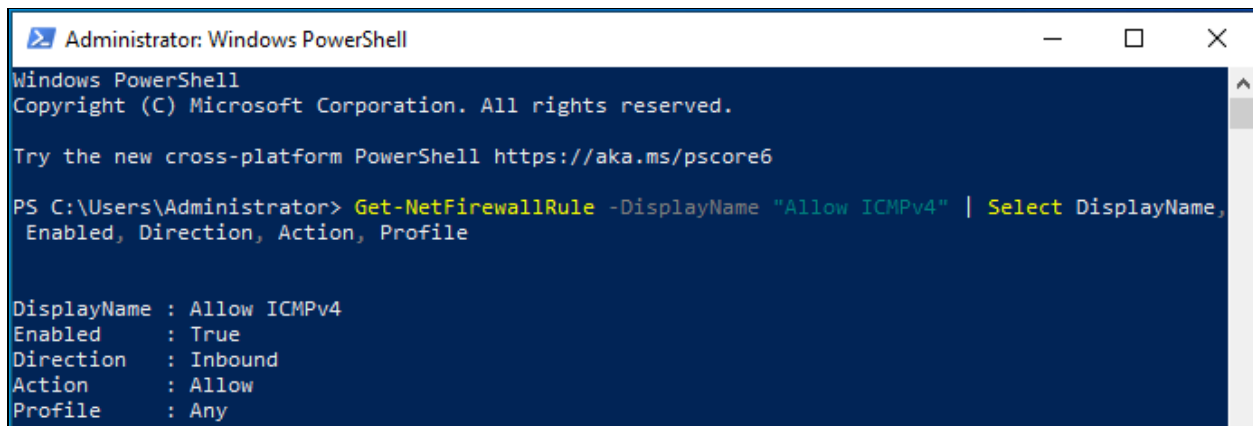
Figure 5. Client IP configuration

3.3 Connectivity and firewall

Objective: confirm the two machines can reach each other, and understand why they could not at first.

Windows Firewall blocks inbound ICMP echo requests by default, so ping fails until a rule allows it. I added an inbound ICMPv4 rule on both machines:

```
New-NetFirewallRule -DisplayName "Allow ICMPv4" -Protocol ICMPv4 -IcmpType 8 -
Action Allow
Get-NetFirewallRule -DisplayName "Allow ICMPv4" | Select DisplayName, Enabled,
Direction, Action, Profile
```



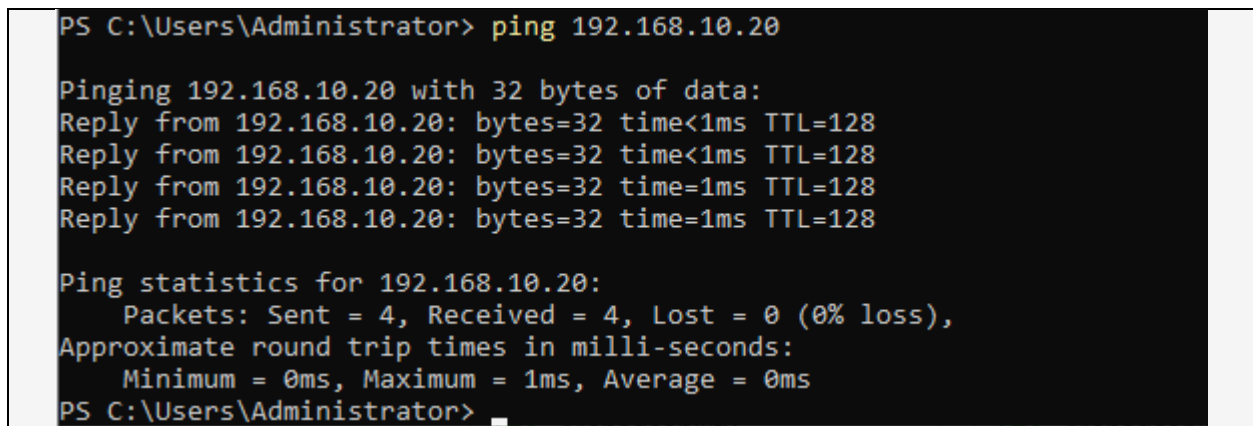
```
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Administrator> Get-NetFirewallRule -DisplayName "Allow ICMPv4" | Select DisplayName,
Enabled, Direction, Action, Profile

DisplayName : Allow ICMPv4
Enabled      : True
Direction   : Inbound
Action       : Allow
Profile      : Any
```

Figure 6. Get-NetFirewallRule output

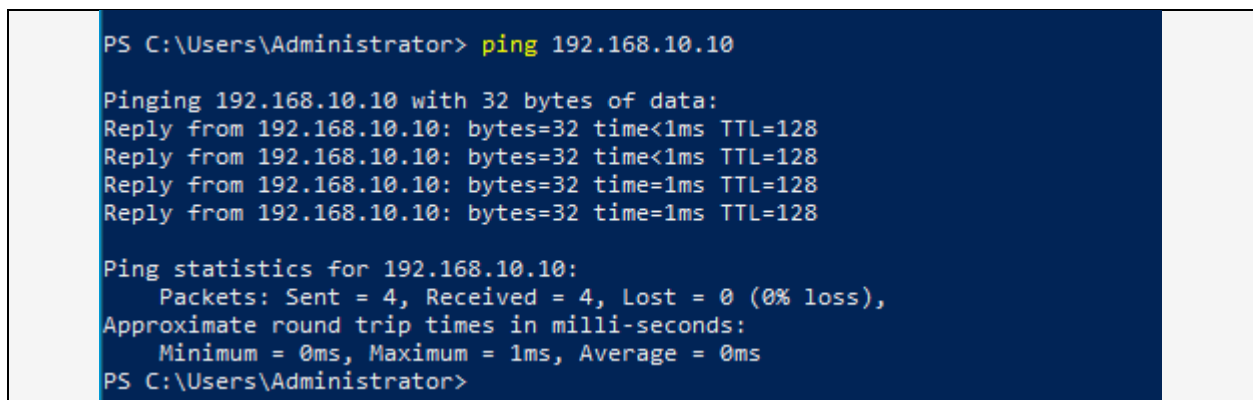


```
PS C:\Users\Administrator> ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time=1ms TTL=128
Reply from 192.168.10.20: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

Figure 7. Ping from the server to the client (0% loss)



```
PS C:\Users\Administrator> ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128
Reply from 192.168.10.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

Figure 8. Ping from the client to the server (0% loss)

3.4 Active Directory Domain Services

Objective: promote the server to a domain controller for a new forest.

1. Server Manager > Manage > Add Roles and Features, and installed the Active Directory Domain Services role.
2. Selected Promote this server to a domain controller > Add a new forest, with root domain lab.local.
3. Kept the Windows Server 2016 functional level, with DNS server and Global Catalog enabled.
4. Set a DSRM (recovery) password and stored it securely.
5. Accepted the NetBIOS name LAB, let the prerequisite check run, installed, and restarted.

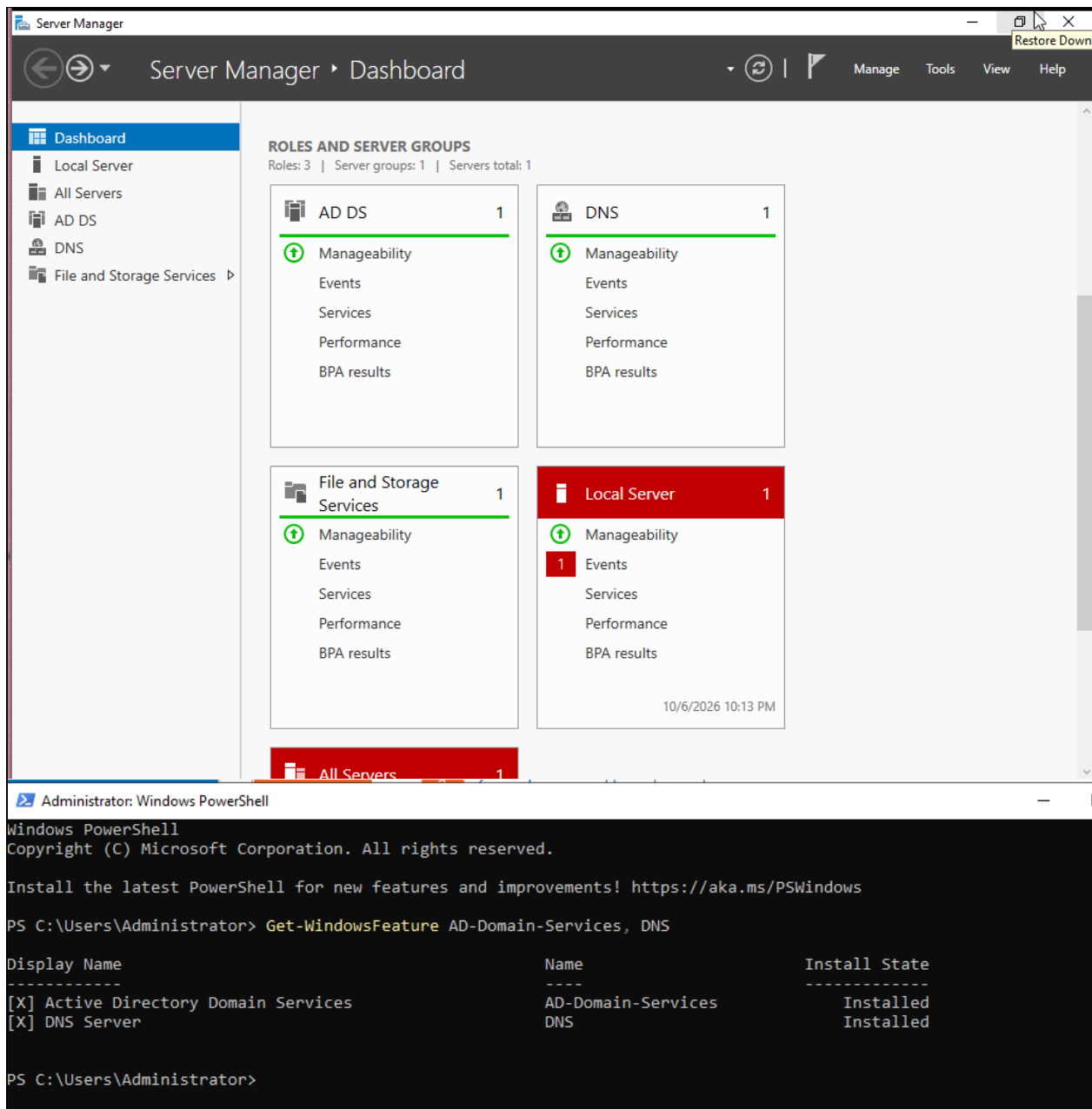


Figure 9. Server Manager showing the AD DS and DNS roles

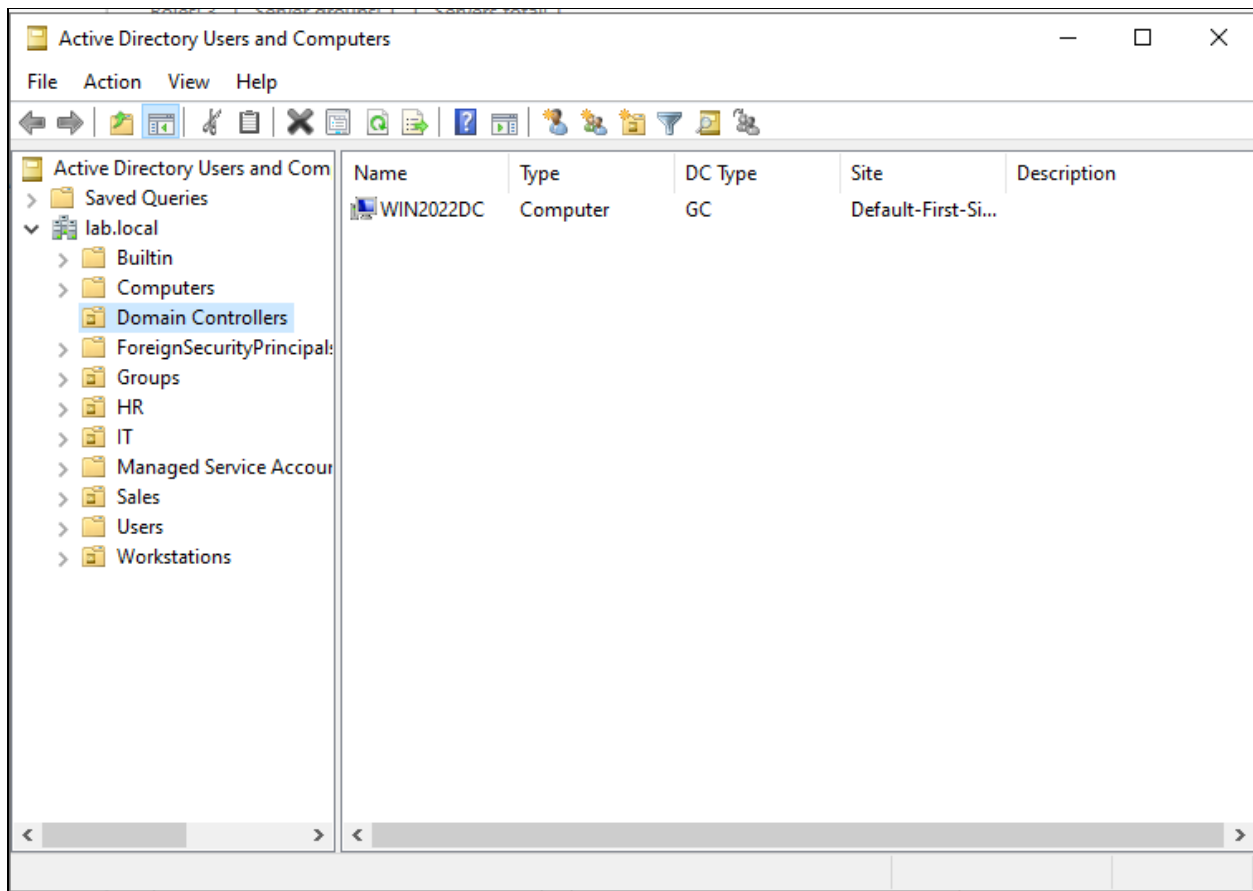


Figure 10. Domain controller promotion summary or the lab.local domain in Active Directory Users and Computers

3.5 DNS verification

Objective: prove the DNS service and the records Active Directory needs are in place.

```
Get-Service DNS
nslookup lab.local
nslookup -type=SRV _ldap._tcp.dc._msdcs.lab.local
```

- Get-Service DNS returned Running.
- nslookup lab.local returned 192.168.10.10.
- DNS Manager showed the lab.local forward lookup zone, the server's A record, and the _msdcs, _tcp, and _udp record folders.
- After promotion, ipconfig /all showed the server using 127.0.0.1 and ::1 for DNS, which is expected for a domain controller that hosts its own DNS.

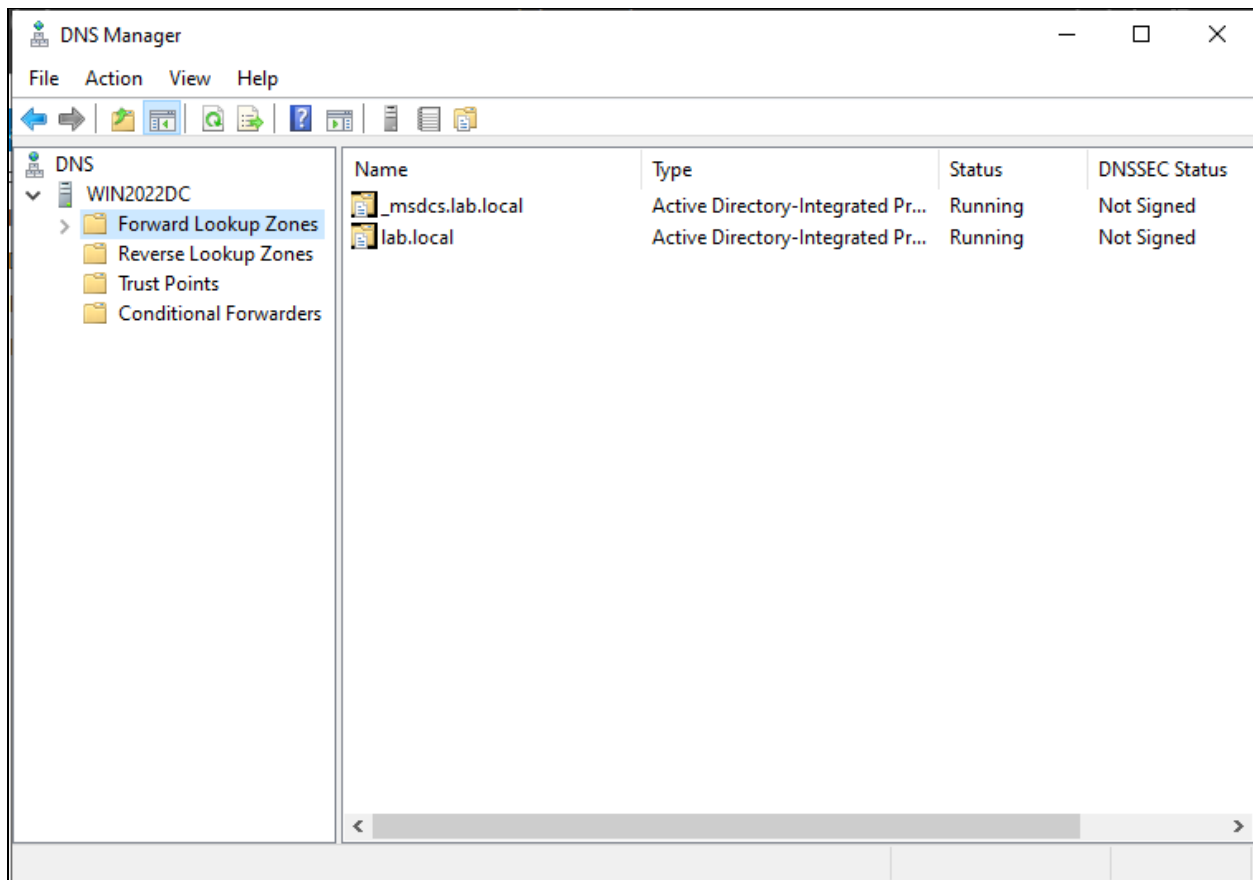


Figure 11. DNS Manager showing the lab.local zone and records

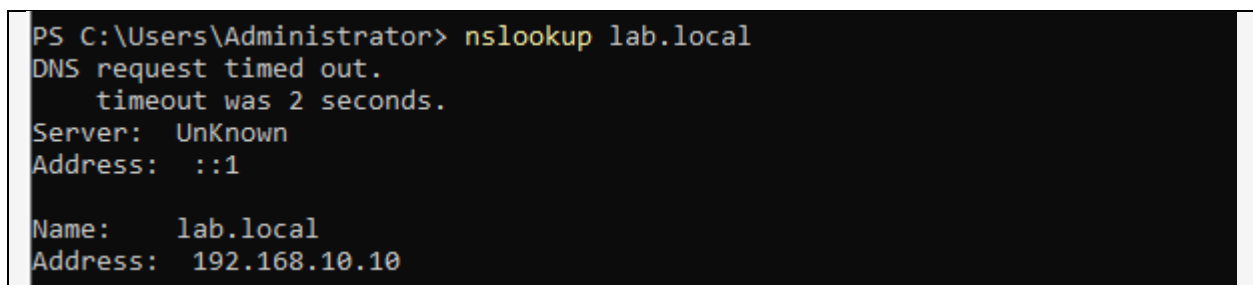


Figure 12. nslookup lab.local output

3.6 Domain join

Objective: join the Windows 10 client to lab.local.

1. Confirmed on the client that ipconfig /all showed DNS 192.168.10.10 and that nslookup lab.local resolved.
2. Ran sysdm.cpl > Computer Name > Change > Domain, entered lab.local, and authenticated as the domain Administrator.
3. Restarted and signed in with a domain account (LAB\username) using Other user on the login screen.
4. Moved the computer object from the default Computers container to the Workstations OU.

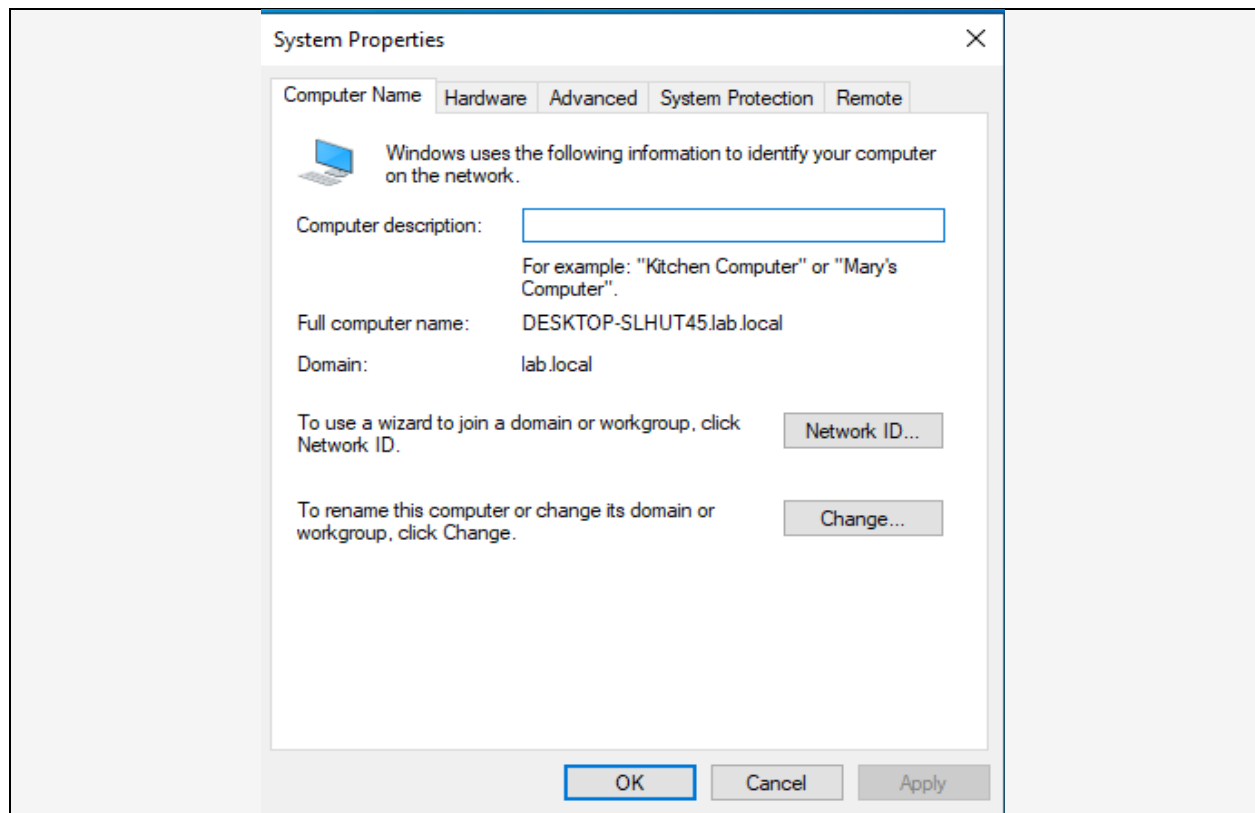


Figure 13. Windows 10 client showing membership in the lab.local domain

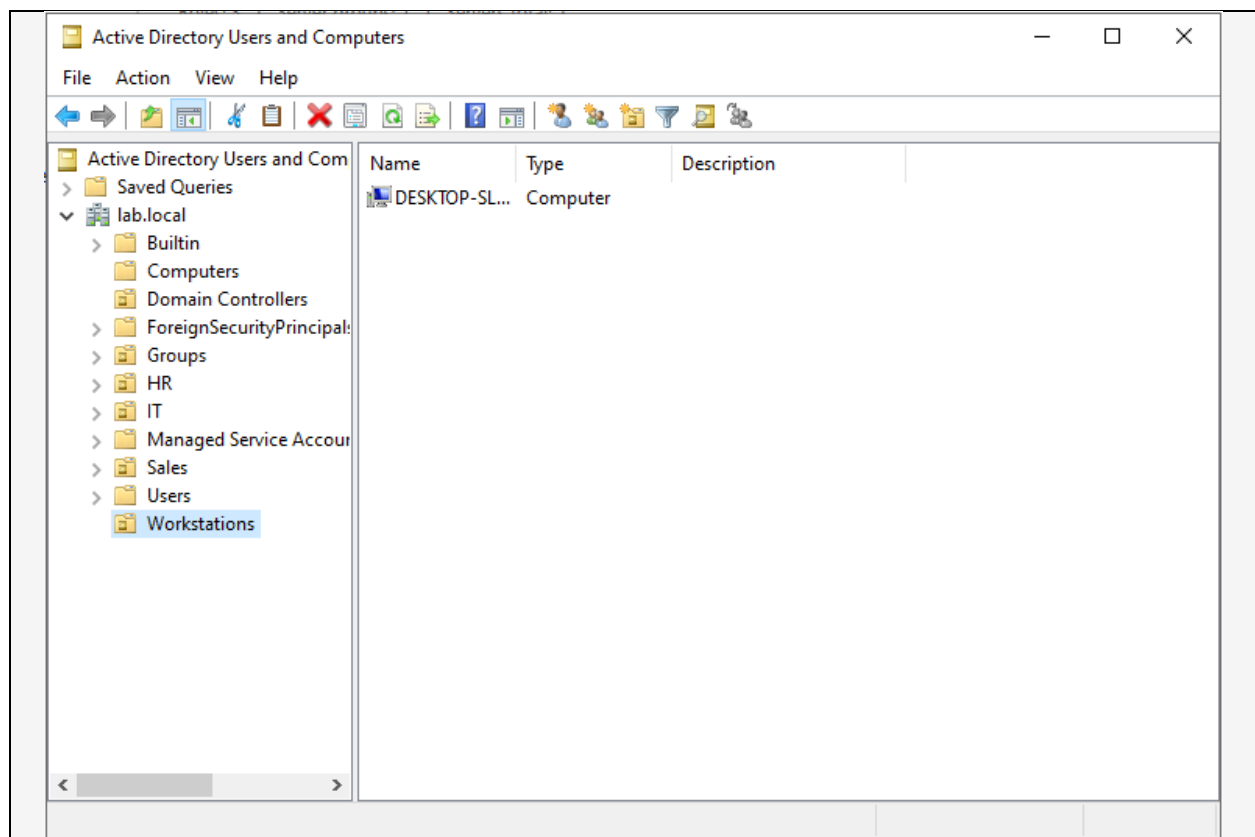


Figure 14. Client computer object in the Workstations OU

3.7 Organizational units

Objective: structure the directory so policy and permissions can be applied by department.

OU	Purpose	Contains
IT	IT department users	a.andrews, b.cooper
HR	HR department users	v.lodge, t.topaz
Sales	Sales department users	k.keller, c.blossom
Workstations	Domain-joined computers, targeted by computer policies	Windows 10 client
Groups	Security groups used for access control	GG-IT, GG-HR, GG-Sales

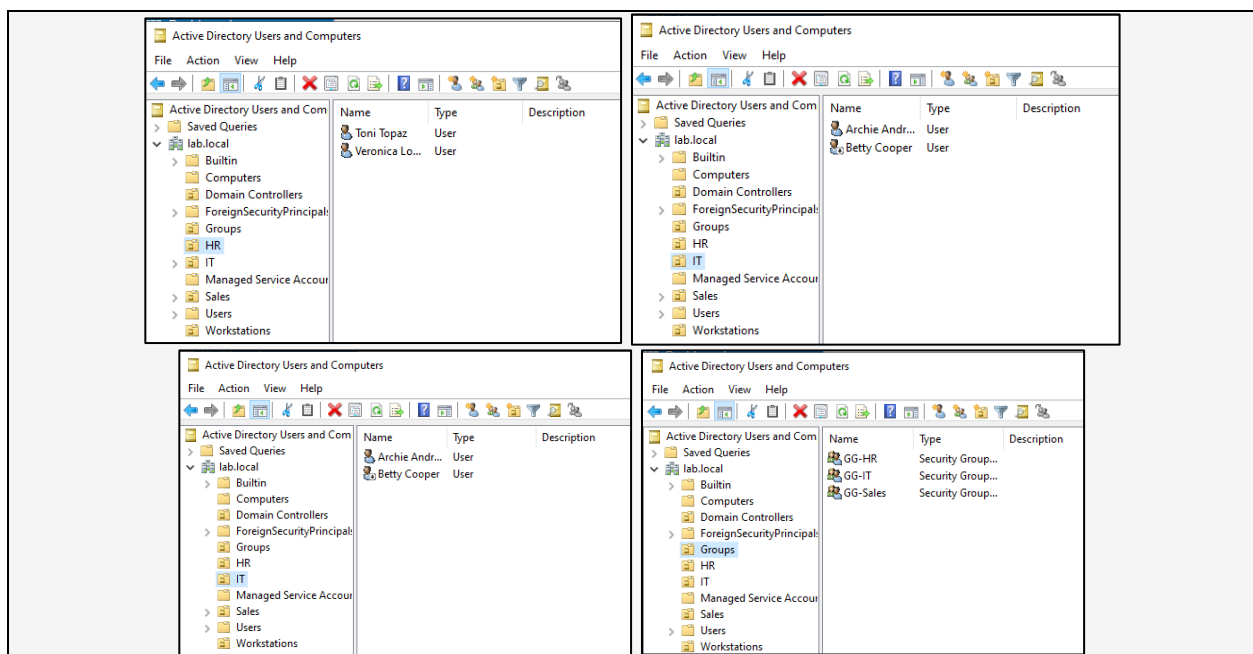


Figure 15. Active Directory Users and Computers with the OUs expanded

3.8 Users and security groups

Objective: create department users and give access through groups, not individual accounts.

Display name	Logon name	OU	Group
Archie Andrews	a.andrews	IT	GG-IT
Betty Cooper	b.cooper	IT	GG-IT
Veronica Lodge	v.lodge	HR	GG-HR
Toni Topaz	t.topaz	HR	GG-HR
Kevin Keller	k.keller	Sales	GG-Sales
Cheryl Blossom	c.blossom	Sales	GG-Sales

Passwords are intentionally not recorded in this document. All accounts are fictional lab accounts.

1. Created each user in the matching department OU (right-click the OU > New > User).
2. Created the global security groups GG-IT, GG-HR, and GG-Sales in the Groups OU.
3. Added each user to the group for their department.
4. Verified membership with the Member Of tab and PowerShell.

```
Get-ADGroupMember "GG-HR" | Select Name, SamAccountName
Get-ADPrincipalGroupMembership v.lodge | Select Name
```

```
PS C:\Users\Administrator> Get-ADGroupMember "GG-HR" | Select Name, SamAccountName

Name                SamAccountName
----                -
Veronica Lodge      v.lodge
Toni Topaz          t.topaz
```

Figure 16. Users inside the department OUs

```
PS C:\Users\Administrator> Get-ADPrincipalGroupMembership v.lodge | Select Name

Name
----
Domain Users
GG-HR
```

Figure 17. Get-ADGroupMember output for the groups

3.9 Group Policy

Objective: enforce settings centrally instead of configuring each machine by hand.

GPO	Linked to	Type	Effect
Default Domain Policy	Domain root	Computer	Password length 12, complexity on, lockout after 5 attempts
Restrict Control Panel	HR, Sales	User	Blocks Control Panel and Settings
Logon Banner	Workstations	Computer	Warning message before logon
Map HR Drive	HR	User preference	Maps \\WIN2022DC\HR as H:
Map IT Drive	IT	User preference	Maps \\WIN2022DC\IT as I:
Map Sales Drive	Sales	User preference	Maps \\WIN2022DC\Sales as S:
Audit Logon Events	Workstations	Computer	Audits logon success and failure, and account lockout failure

Password and account lockout policy

Setting	Value
Minimum password length	12 characters
Password must meet complexity requirements	Enabled
Account lockout threshold	5 invalid attempts
Account lockout duration	30 minutes
Reset account lockout counter after	30 minutes

Set in the Default Domain Policy, because account policies for domain users only take effect from a GPO linked at the domain root. Verification:

```
Get-ADDefaultDomainPasswordPolicy
gpupdate /force
net accounts /domain
```

Negative test: I attempted to reset a user's password to a short value and Windows rejected it.

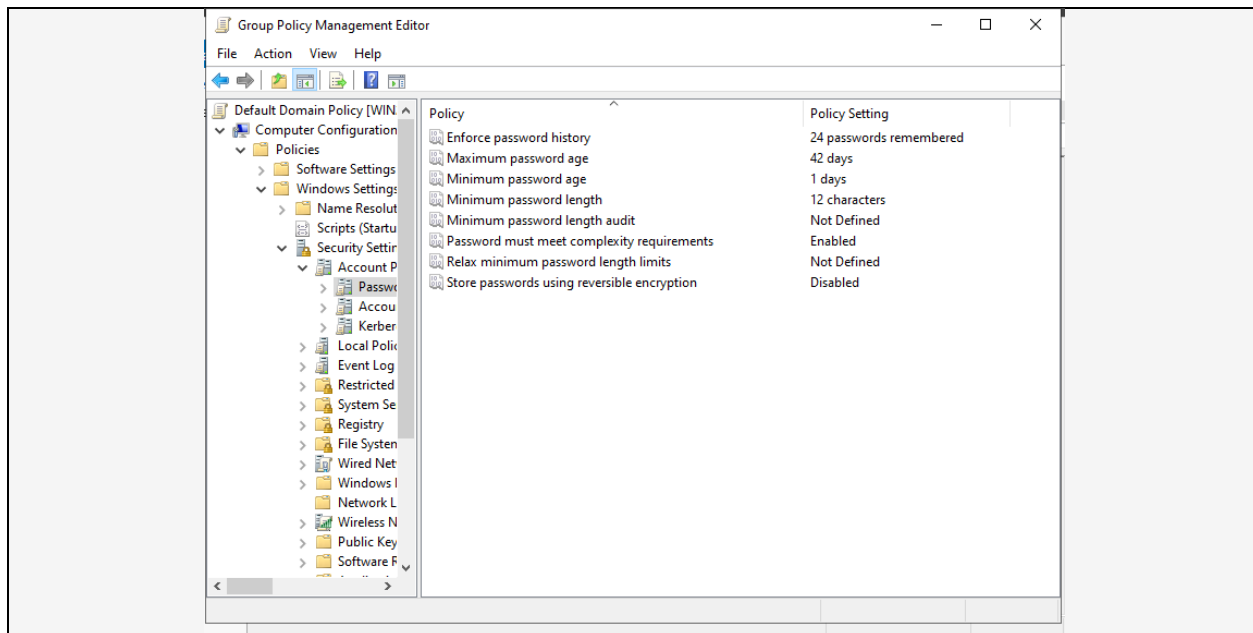


Figure 18. Password policy settings in the Group Policy editor

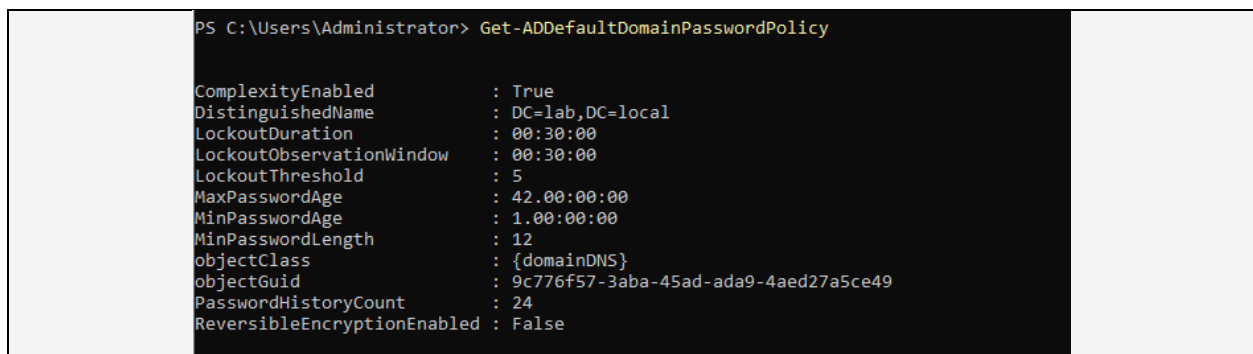


Figure 19. Get-ADDefaultDomainPasswordPolicy or net accounts /domain output

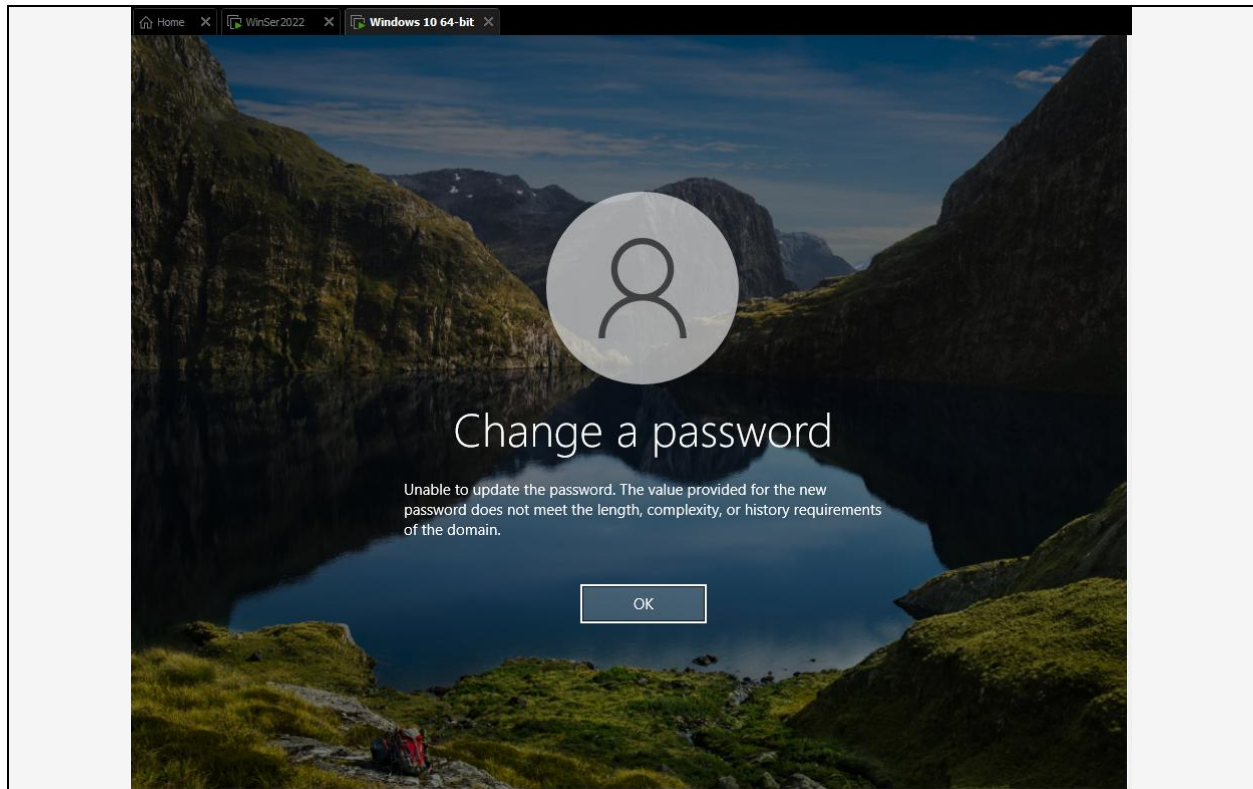


Figure 20. Error message when setting a short password

Restrict Control Panel

Setting: User Configuration > Policies > Administrative Templates > Control Panel > Prohibit access to Control Panel and PC settings (Enabled). Linked to the HR and Sales OUs, because it is a user policy and applies to the users inside those OUs.

Test	Expected	Result
HR user opens Control Panel (control)	Blocked by restriction message	Pass
IT user opens Control Panel	Opens normally	Pass

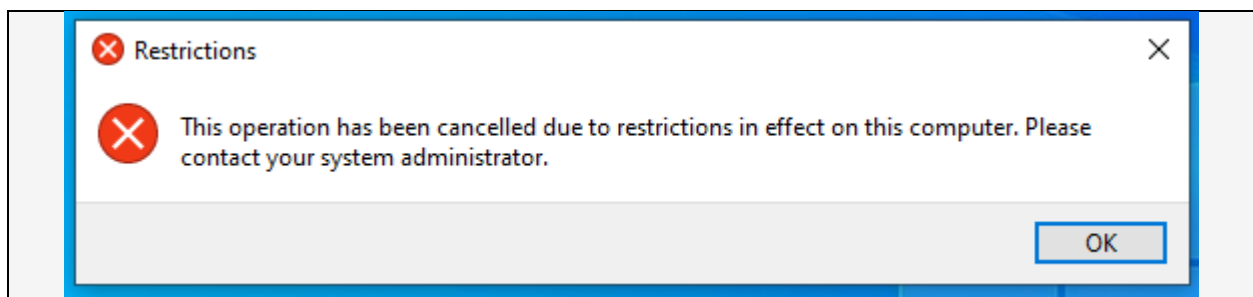


Figure 21. Restriction message for an HR or Sales user

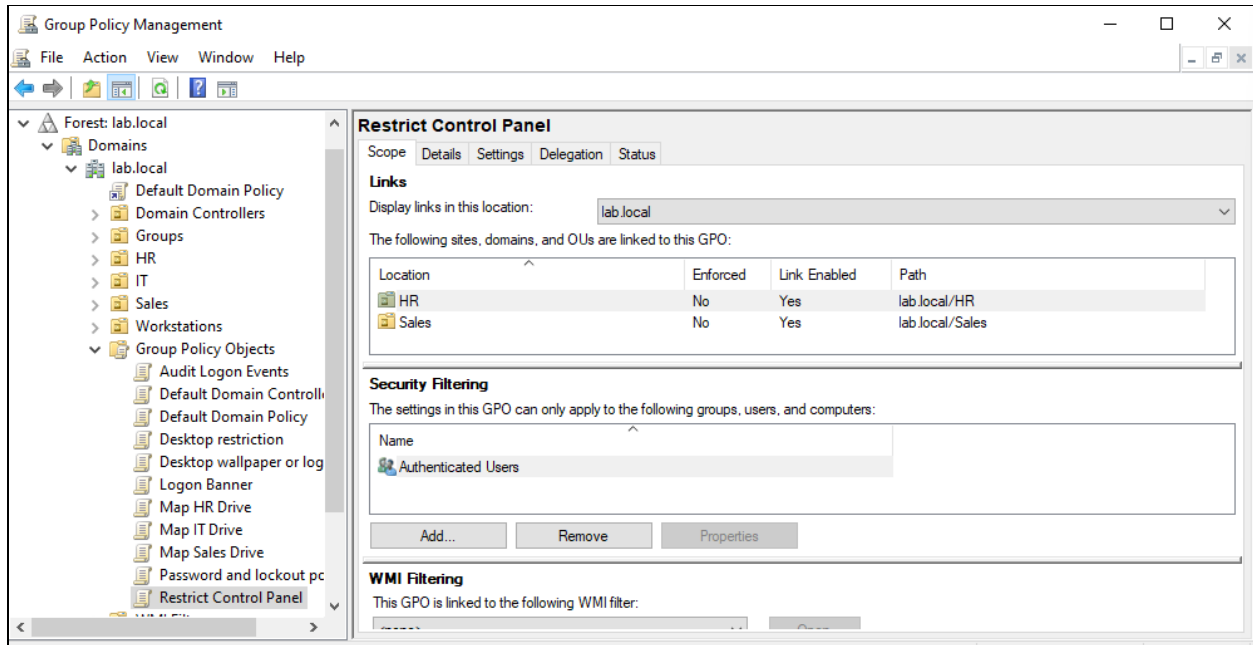


Figure 22. GPO linked to the HR and Sales OUs in Group Policy Management

Logon banner

Setting: Computer Configuration > Windows Settings > Security Settings > Local Policies > Security Options, with both the message title and message text defined. Linked to the Workstations OU.

Field	Value
Title	Authorized Use Only
Text	This system is for authorized users only. Activity may be monitored and logged. Unauthorized access is prohibited.

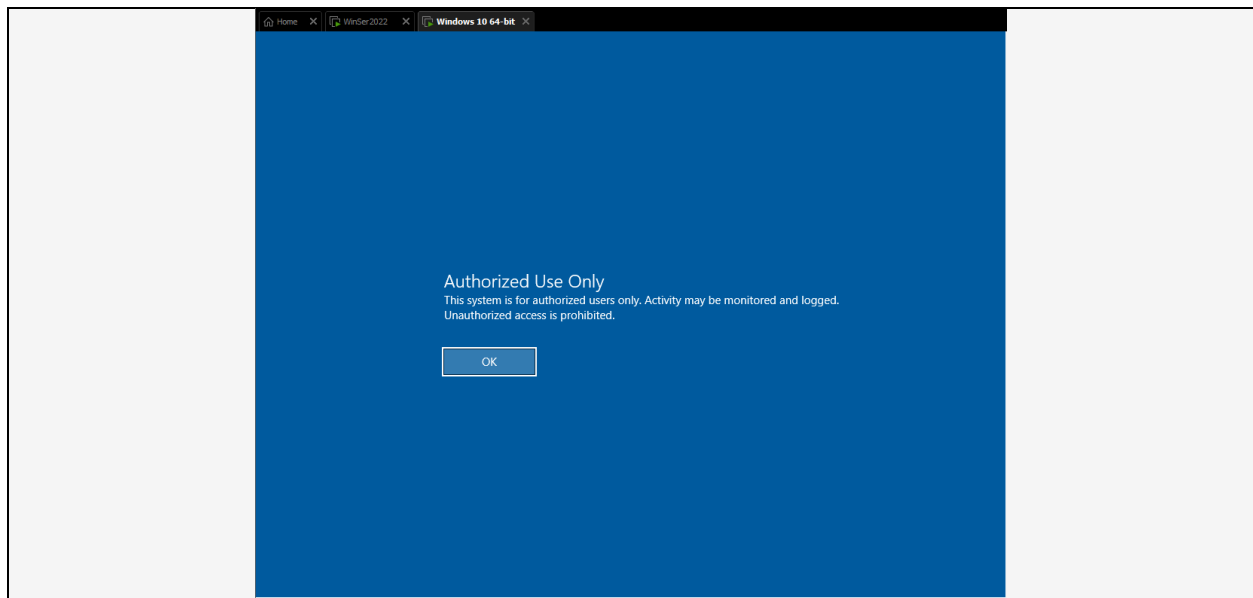


Figure 23. Logon banner on the Windows 10 client

Mapped drives

Setting: User Configuration > Preferences > Windows Settings > Drive Maps, with Action set to Create and Reconnect ticked. Each department gets only its own drive.

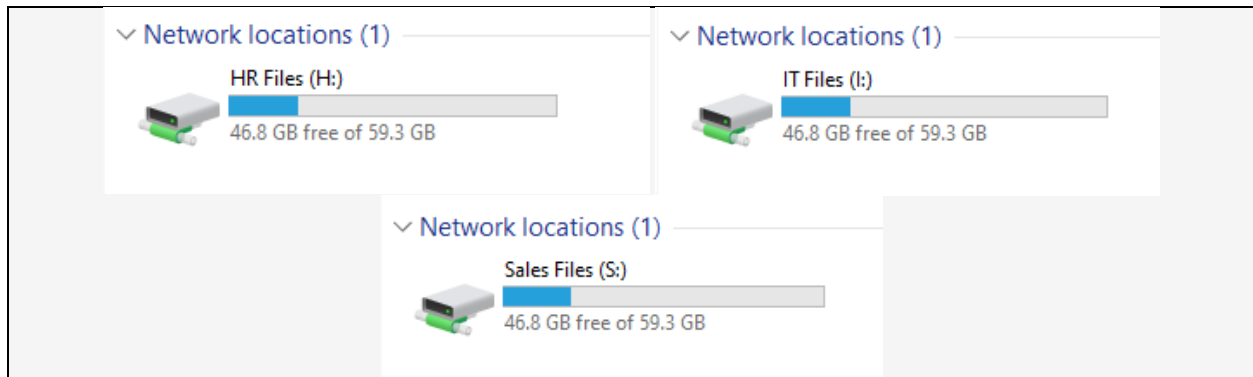


Figure 24. This PC on the client showing the mapped network drive

Audit policy

Setting: Advanced Audit Policy Configuration > Logon/Logoff > Audit Logon (Success and Failure) and Audit Account Lockout (Failure). Verified on the client with:

```
gpupdate /force
auditpol /get /category:"Logon/Logoff"
```

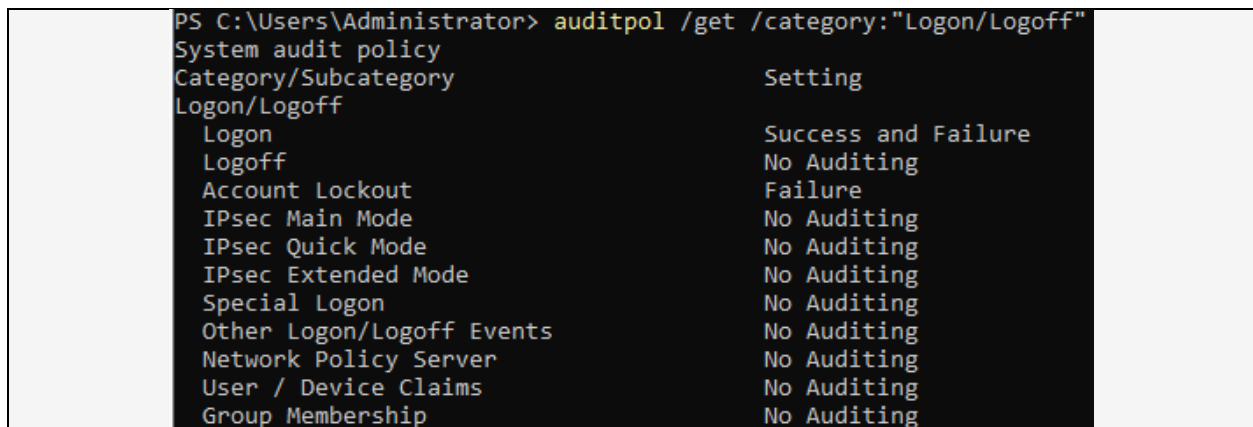


Figure 25. auditpol output

Verifying applied policy

```
gpresult /r /scope user
gpresult /r /scope computer
```

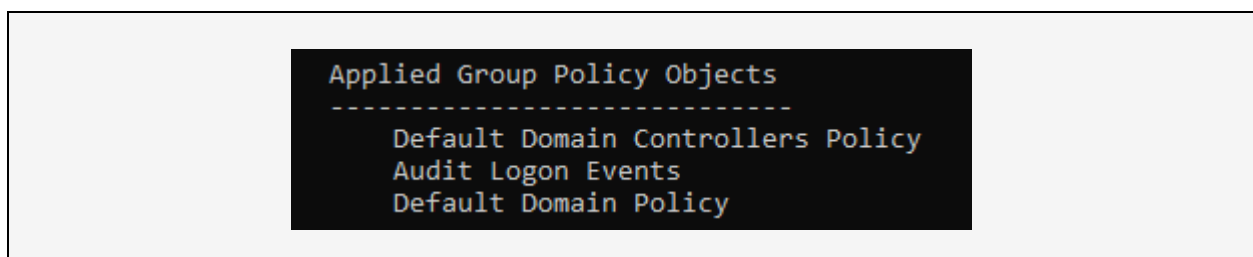


Figure 26. gpresult output listing the applied GPOs

3.10 File shares and NTFS permissions

Objective: give each department access to only its own folder.

Folder	Share name	Share permission	NTFS permission
C:\Shares\HR	HR	Everyone: Full Control	GG-HR: Modify; Administrators and SYSTEM: Full Control
C:\Shares\IT	IT	Everyone: Full Control	GG-IT: Modify; Administrators and SYSTEM: Full Control
C:\Shares\Sales	Sales	Everyone: Full Control	GG-Sales: Modify; Administrators and SYSTEM: Full Control

The share permission is left open and NTFS does the restricting, so access is managed in one place. The effective permission is the more restrictive of the two.

1. Created the folders under C:\Shares and shared each one with Advanced Sharing.
2. On each subfolder, disabled inheritance (converting inherited permissions to explicit), removed the Users entries, and added the department group with Modify.
3. Mapped each department's folder through its drive-map GPO.

```
Get-SmbShare
icacls C:\Shares\HR
icacls C:\Shares\IT
icacls C:\Shares\Sales
```

Test	Expected	Result
HR user creates a file on the HR drive	Succeeds	Pass
HR user opens \\WIN2022DC\IT	Access denied	Pass
HR user opens \\WIN2022DC\Sales	Access denied	Pass
IT user opens the IT share	Succeeds	Pass
IT user opens \\WIN2022DC\HR	Access denied	Pass

PS C:\Users\Administrator> Get-SmbShare			
Name	ScopeName	Path	Description
ADMIN\$	*	C:\Windows	Remote Admin
C\$	*	C:\	Default share
HR	*	C:\Shares	
IPC\$	*		Remote IPC
IT	*	C:\Shares	
NETLOGON	*	C:\Windows\SYSVOL\sysvol\lab.local\SCRIPTS	Logon server share
Sales	*	C:\Shares	
SYSVOL	*	C:\Windows\SYSVOL\sysvol	Logon server share

Figure 27. Get-SmbShare output

```

PS C:\Users\Administrator> icacls C:\Shares\HR
C:\Shares\HR NT AUTHORITY\SYSTEM:(OI)(CI)(F)
BUILTIN\Administrators:(OI)(CI)(F)
CREATOR OWNER:(OI)(CI)(IO)(F)
LAB\GG-HR:(OI)(CI)(M)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> icacls C:\Shares\IT
C:\Shares\IT NT AUTHORITY\SYSTEM:(OI)(CI)(F)
BUILTIN\Administrators:(OI)(CI)(F)
CREATOR OWNER:(OI)(CI)(IO)(F)
LAB\GG-IT:(OI)(CI)(M)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\Administrator> icacls C:\Shares\Sales
C:\Shares\Sales NT AUTHORITY\SYSTEM:(OI)(CI)(F)
BUILTIN\Administrators:(OI)(CI)(F)
CREATOR OWNER:(OI)(CI)(IO)(F)
LAB\GG-Sales:(OI)(CI)(M)

Successfully processed 1 files; Failed processing 0 files

```

Figure 28. icacls output for the three folders

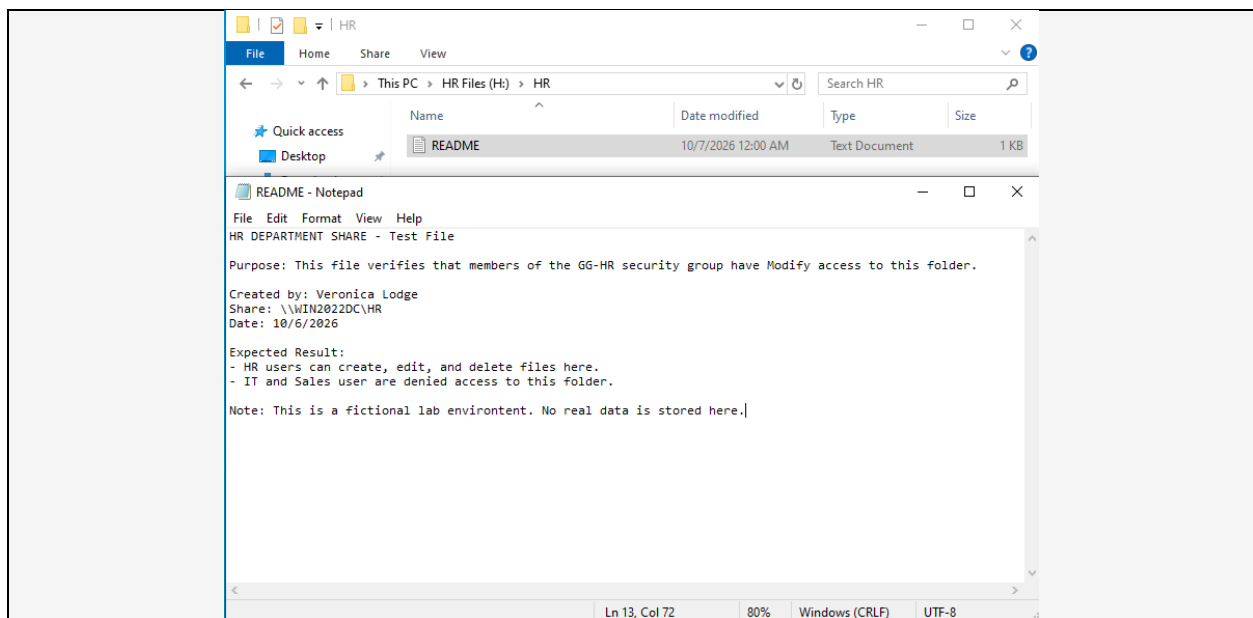


Figure 29. HR user creating a file on the HR drive

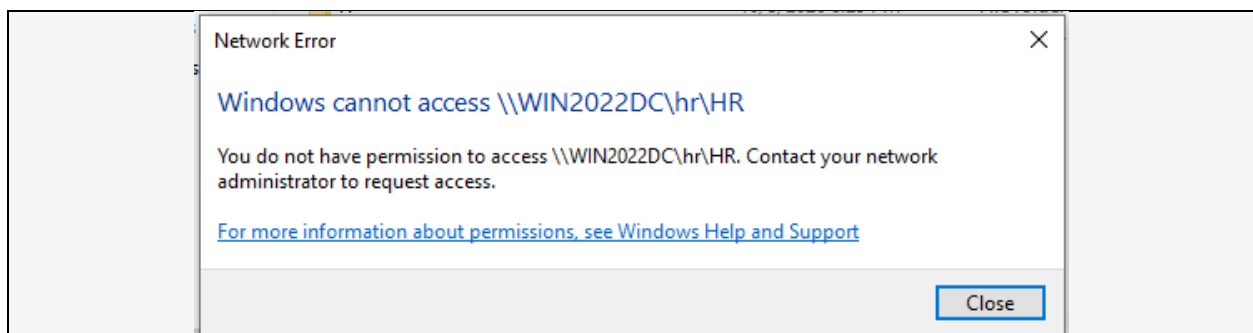


Figure 30. Access denied message on another department's share

3.11 Account lockout and event log analysis

Objective: generate a lockout on purpose and trace it through the logs.

1. On the client, entered a wrong password for a test user five times at the login screen (Other user > LAB\username).
2. On the sixth attempt, Windows reported the account was locked out.
3. On the domain controller, filtered the Security log for Event ID 4740.
4. On the client, filtered the Security log for Event ID 4625 to see the individual failed logons.
5. Unlocked the account in Active Directory Users and Computers (Account tab > Unlock account).

Event ID	Where	Meaning
4625	Client	A logon attempt failed
4740	Domain controller	An account was locked out; shows the account and the source computer
4767	Domain controller	An account was unlocked
4771 / 4776	Domain controller	Failed Kerberos or credential validation attempts

Analysis of a captured Event 4625

Field	Value	Meaning
Account name	a.andrews (domain LAB)	The account whose logon failed
Logon type	2	Interactive: typed at the machine's login screen
Status / Sub status	0xC000006D / 0xC000006A	Logon failed because the password was wrong for an existing account (an unknown user shows 0xC0000064)
Subject	SYSTEM, computer account	The computer's own logon process handled the attempt; this is normal for login-screen failures
Logged	10/6/2026 7:37:30 AM	When it occurred

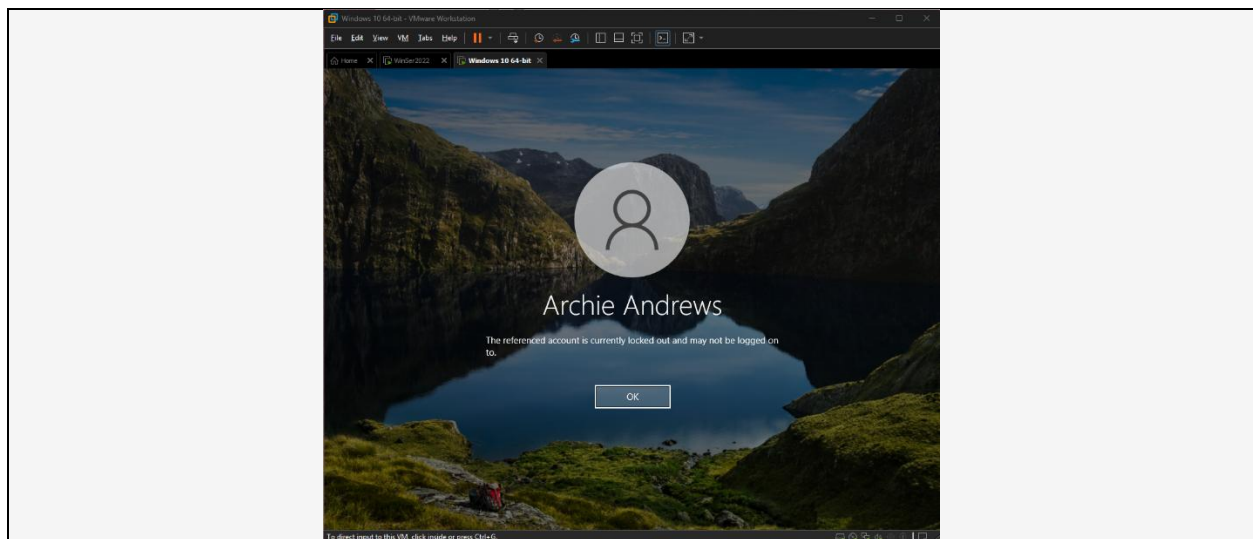


Figure 31. Lockout message on the client

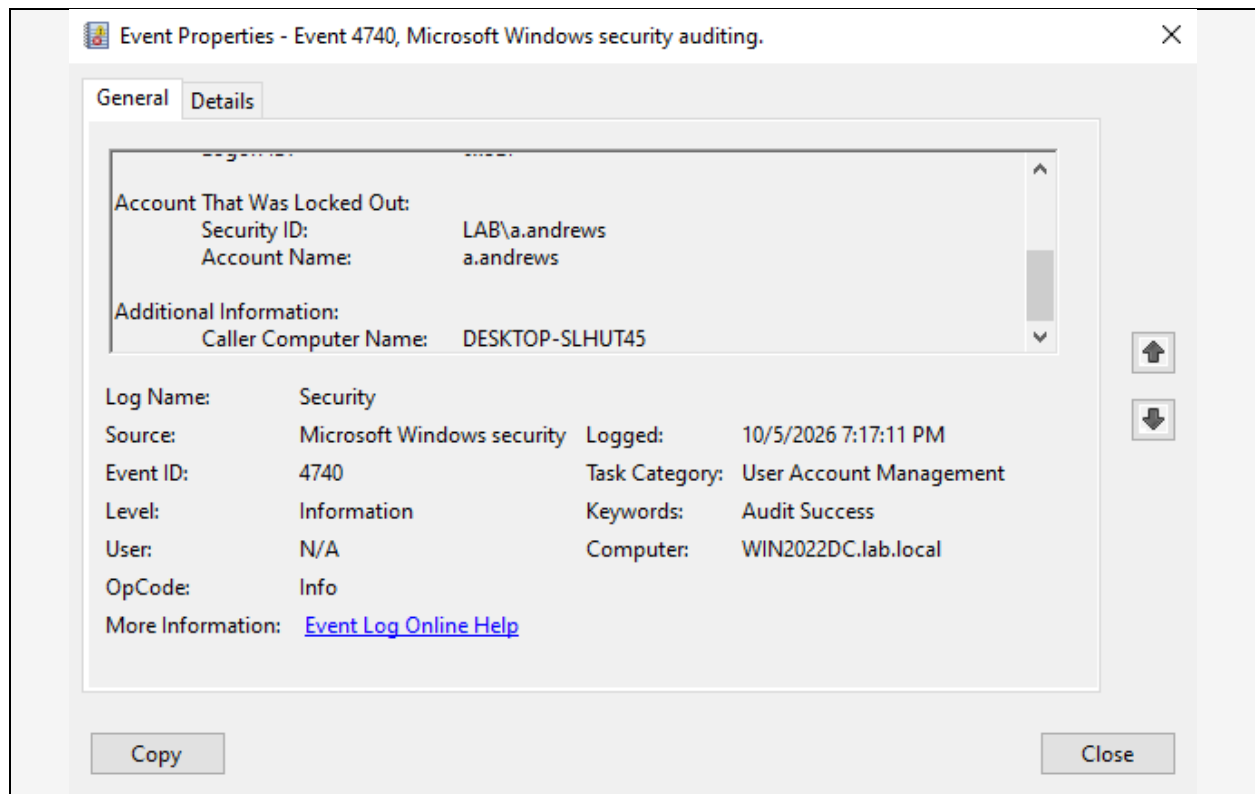


Figure 32. Event ID 4740 details on the domain controller

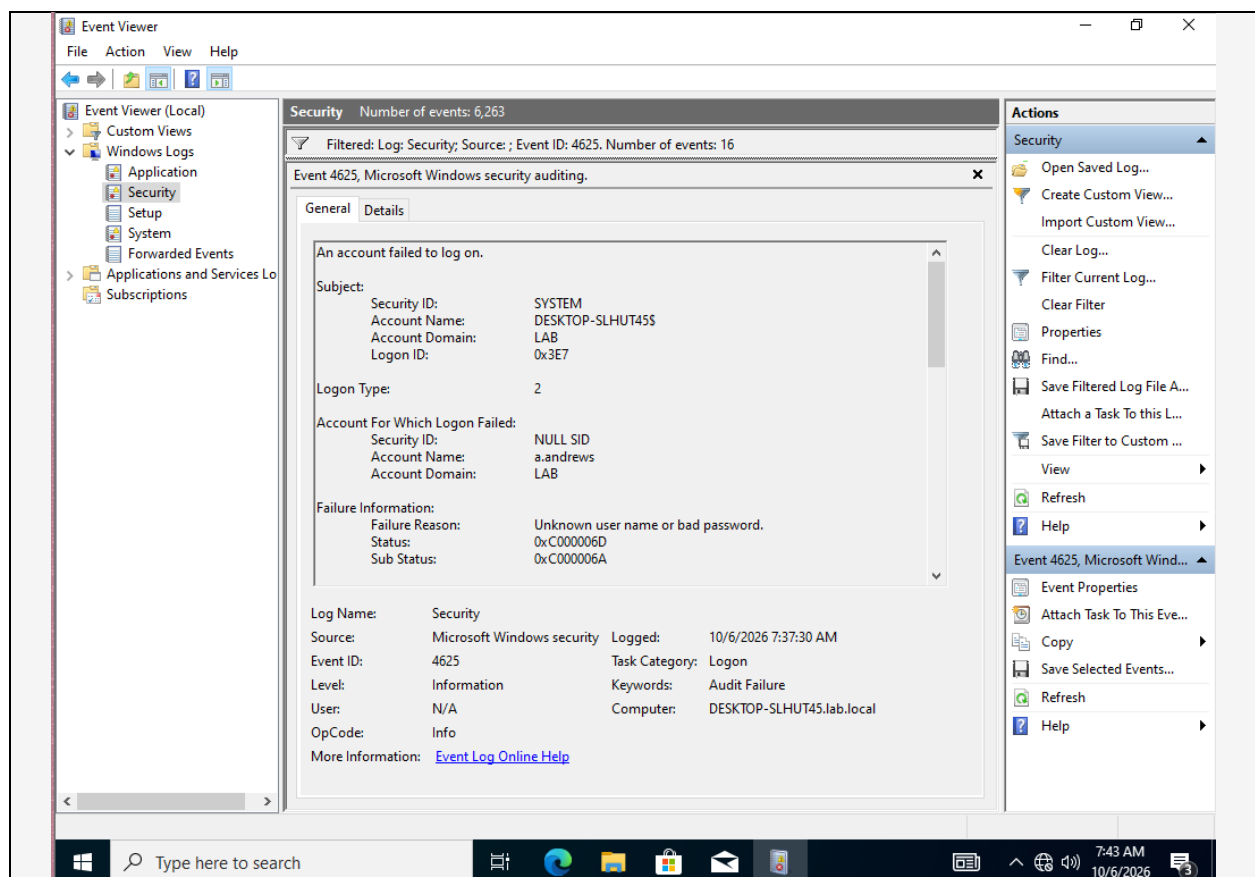


Figure 33. Event ID 4625 details on the client

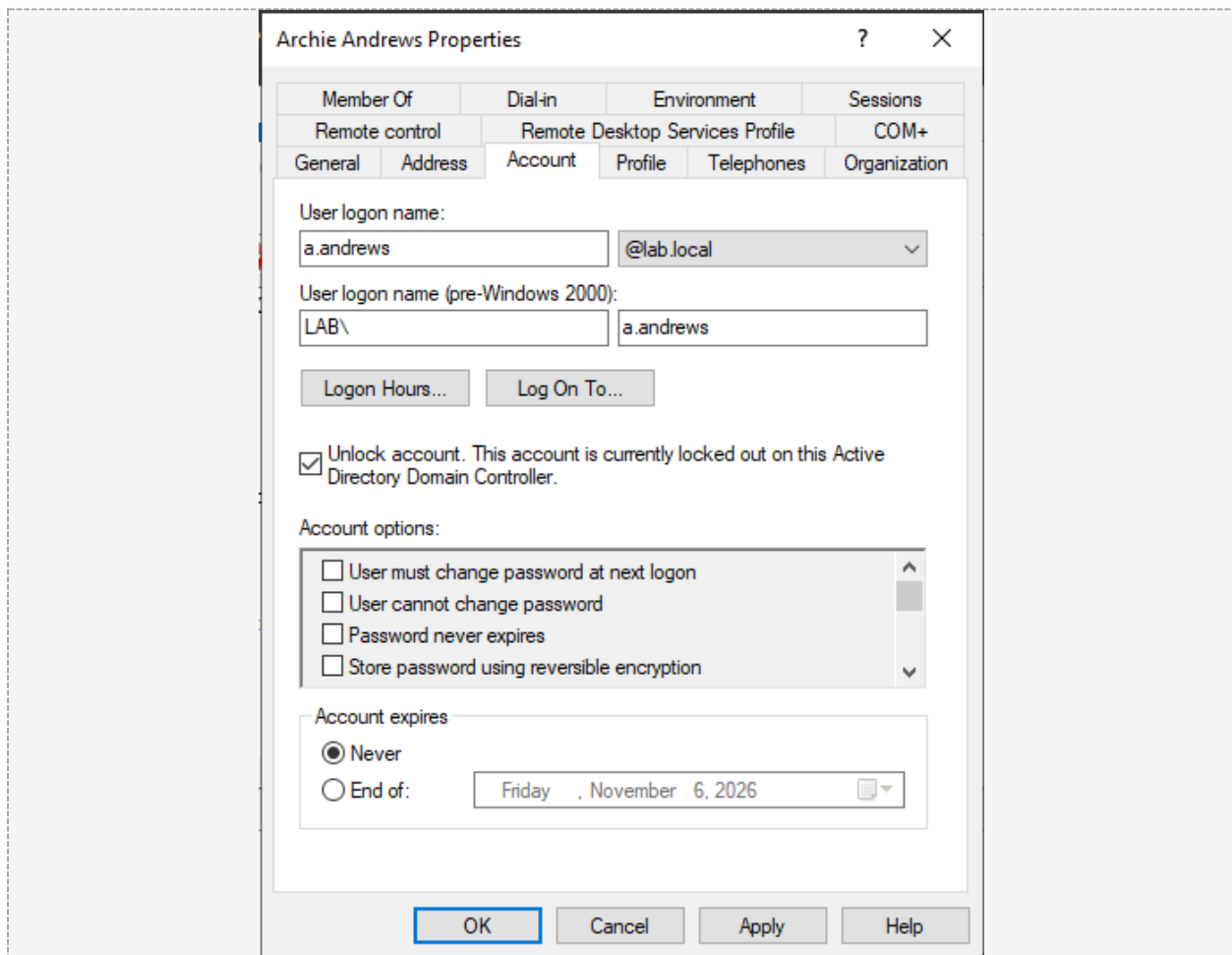


Figure 34. Unlock account in Active Directory Users and Computers

4. Test results summary

#	Test	Expected result	Result
1	Ping server to client and client to server	Replies, 0% loss	Pass
2	nslookup lab.local from the client	Returns 192.168.10.10	Pass
3	Domain join and domain login	Welcome message, login works	Pass
4	Group membership check	Each user in the correct group	Pass
5	Password policy values	Length 12, lockout 5	Pass
6	Short password rejected	Error from Windows	Pass
7	Control Panel blocked for HR/Sales, allowed for IT	As described	Pass
8	Logon banner appears	Banner before login	Pass
9	Mapped drive appears for the department user	Drive visible	Pass
10	Access to own share / denied on others	Allowed / denied	Pass
11	Account lockout after 5 bad passwords	Locked, Event 4740 logged	Pass
12	Event 4625 analysis	Logon type and sub status identified	Pass

5. Troubleshooting log

Record only problems you actually saw. Real problems, even small ones, are the best material for an interview.

Symptom	Diagnosis	Cause and fix
Mapped drive did not appear for a domain user	Checked the signed-in account with whoami in a normal prompt (an elevated prompt reports the admin account), reviewed gpresult /r /scope user, and checked the GPO existed and was linked to the HR OU	The drive-mapping GPO had not been created yet, so there was no policy to apply to the HR user; created the GPO, linked it to the HR OU, and added a drive map.
Ping blocked between the VMs	Checked the target machine's firewall rule, adapter state, and subnet	Inbound ICMPv4 rule was missing; added it on both machines
Shutdown Event Tracker prompt on the server	Read the dialog and event log	The VM had been powered off instead of shut down from inside Windows; a clean shutdown avoids it
Server DNS changed to 127.0.0.1 after promotion	Compared ipconfig /all with the earlier configuration and ran nslookup	Expected for a domain controller that hosts its own DNS; no change needed

Symptom	Diagnosis	Cause and fix
UAC prompt asked for credentials when opening an elevated Command Prompt on the client	Read the prompt, which named Windows Command Processor with Domain: LAB, and checked which account was signed in	A standard domain user has no admin rights, so elevation needs administrator credentials. Entered the domain Administrator credentials for tasks that needed elevation, and used a normal prompt for user-level checks such as <code>gpresult /r /scope user</code>
VMnet1's subnet (192.168.159.0) did not match the planned lab range (192.168.10.0/24)	Compared the Virtual Network Editor with <code>ipconfig /all</code> on both VMs	VMware's default host-only subnet differed from my addressing plan; changed VMnet1 to 192.168.10.0/24 and kept its DHCP service off

6. Lessons learned, limitations, and next steps

What I learned

- Building this lab taught me how closely Active Directory depends on DNS. The Windows 10 client could only find and join lab.local because its DNS pointed to the domain controller, which stores the records used to locate it, and checking `nslookup` is now the first thing I do when a domain problem appears. I also learned that groups scale better than individual permissions: I gave each department folder's NTFS permissions to a security group instead of to users, so moving someone between departments only means changing their group, and testing both an allowed case and a denied case showed the restriction really worked. Group Policy taught me that where a policy is linked decides who it affects. User policies such as the Control Panel restriction and drive maps follow the user's OU, computer policies such as the logon banner follow the computer's OU, and when a drive map did not appear I traced the cause with `gpresult` instead of guessing. Reading the event logs showed me how much detail a single entry holds: Event 4625 records each failed logon, with the logon type and sub-status separating a wrong password from an unknown user, while Event 4740 on the domain controller records the lockout that followed and the computer it came from. Finally, I learned to verify before and after every change, whether that meant reading the folder name before editing permissions, checking which account a prompt was really running as, or remembering that a failed ping can mean a firewall rule is missing and not that the network is down.

Limitations

- Single domain controller and one client, so there is no replication or redundancy.
- Addresses are static; there is no DHCP server, routing, or VLANs in this lab.
- Baseline hardening only (no LAPS, tiered administration, or patch management), and logs are read manually in Event Viewer.

Possible next steps

- DHCP server with a scope and reservations
- Second domain controller and replication
- Central log collection with a SIEM such as Wazuh
- Delegated administration and PowerShell user provisioning

Appendix A. Command reference

Command	Purpose
ipconfig /all	Show IP, mask, gateway, DNS, and DHCP state
ping <address>	Test reachability (needs an ICMP rule on the target)
nslookup lab.local	Test DNS resolution and see which DNS server answered
Get-Service DNS	Check that the DNS service is running
Get-NetFirewallRule	Check a firewall rule's state and scope
Get-ADGroupMember	List the members of a group
Get-ADDefaultDomainPasswordPolicy	Show the domain password and lockout policy
net accounts /domain	Show password policy as the client sees it
gpupdate /force	Refresh Group Policy immediately
gpresult /r	List the GPOs applied to a user or computer
auditpol /get /category:*	Show audit policy settings
Get-SmbShare	List shared folders
icacls <folder>	Show or set NTFS permissions
whoami	Show the account a prompt is running as

Appendix B. Concepts I can explain

Concept	One-line explanation
Subnet mask	Splits an address into the network part and the host part; /24 means the first three numbers identify the network
Default gateway	The router used to reach other networks; not needed on an isolated subnet
Why DNS matters for AD	Clients find domain controllers through SRV records stored in DNS
Static IP vs DHCP	Servers need fixed addresses; clients can use DHCP
OU vs group	OUs organize objects and receive GPOs; groups receive permissions
Share vs NTFS permissions	Share controls network access, NTFS controls file access; the stricter one wins
User vs computer policy	User policies follow the user's OU; computer policies follow the computer's OU
Event 4625 vs 4740	4625 is a single failed logon; 4740 is the account lockout that results from repeated failures

Appendix C. Security notes

- All accounts and passwords are fictional lab credentials and are not recorded here.
- Do not reuse lab passwords for real accounts.
- The lab network is isolated (host-only) and has no internet access.